TASK 2: STUDY MULTI-AGENT AI SYSTEMS

Project: AI-Based Startup Idea Validator with Market Analysis Assistance

1. Understand AI Agents and LLMs

AI Agent: An AI agent is a system that performs a specific task, such as searching for information or analysing competitors.

LLM (Large Language Model): An AI model that understands and generates human language. It can summarise information, answer questions, and generate reports.

Example: An AI agent can use an LLM to analyse a startup idea and provide suggestions.

2. How Multiple AI Agents Work Together

Multiple AI agents divide a big task into smaller tasks. Each agent performs its assigned work, and the results are combined to generate a final output.

Example: For a food delivery startup idea, one agent researches the market, another analyses competitors, and another identifies risks. Finally, the report agent combines the results.

3. Responsibilities of the Six Agents

1. Web Search Agent

- Searches the internet for information related to the startup idea.
- Collects industry trends, customer needs, and information about existing businesses.
- Saves useful source links for verification.

2. Market Analysis Agent

- Studies the target market and potential customers.
- Identifies market demand, customer segments, and trends.
- Helps understand whether the startup idea may have market opportunities.

3. Competitor Analysis Agent

- Identifies businesses offering similar products or services.
- Compares features, pricing, strengths, and weaknesses where information is available.
- Identifies opportunities to make the startup different from competitors.

4. SWOT and Risk Analysis Agent

- SWOT means Strengths, Weaknesses, Opportunities, and Threats.
- Identifies advantages, areas to improve, growth opportunities, and external challenges.
- Highlights risks such as high costs, low customer demand, and operational difficulties.

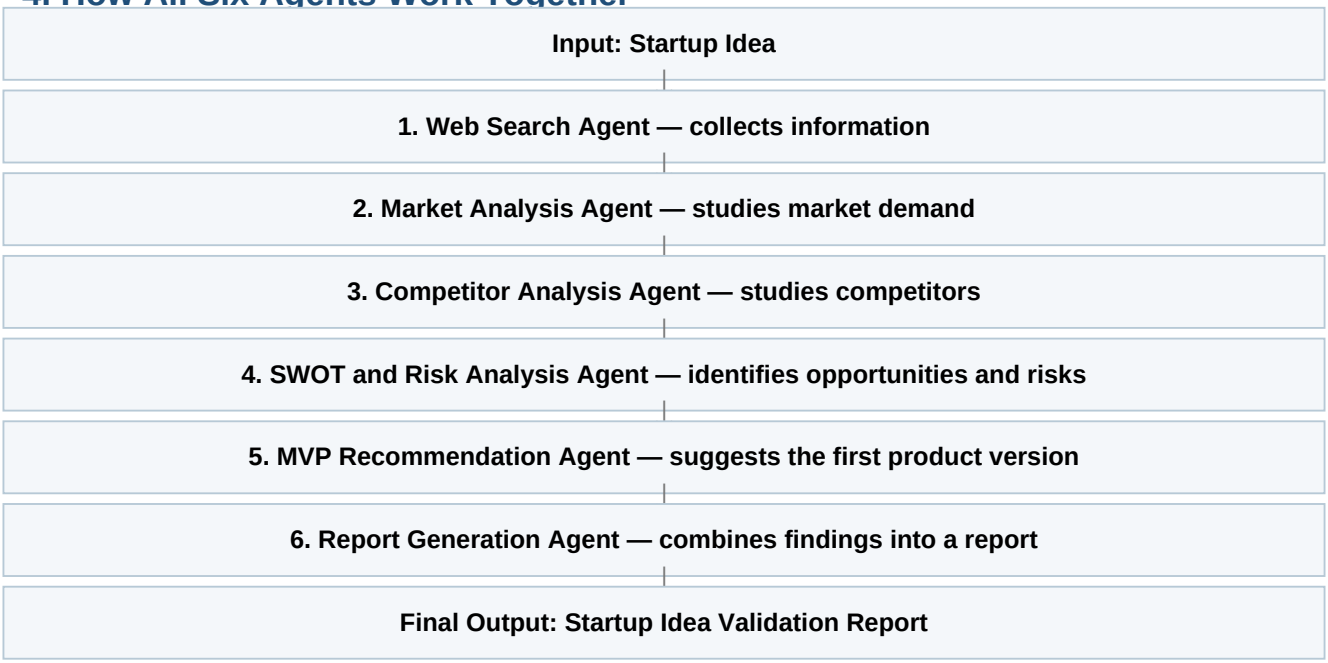
5. MVP Recommendation Agent

- Suggests essential features for the first version of the product.
- Recommends a simple way to test the startup idea before investing heavily.
- Helps identify important features based on customer needs.

6. Report Generation Agent

- Collects outputs from all the other agents.
- Organises the information into a structured report.
- Presents market findings, competitor analysis, SWOT analysis, risks, and recommendations.

4. How All Six Agents Work Together



5. Conclusion

Multi-agent AI systems divide complex tasks among specialised agents. In our Startup Idea Validator project, six agents work together to collect information, analyse the market and competitors, identify risks, recommend an MVP, and generate a final report. This helps entrepreneurs evaluate startup ideas in a structured way.
AI-generated findings should be checked against reliable sources before making business decisions.